

ADVANCED PRINCIPLES OF MECHANICS I

NOTES ON USC SPRING 2026 COURSE PHYS-161

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Special Relativity

SECTION 1

Dimensions and Dimensional Analysis

This will be an experimental course: starting with the concept of space-time, going back to kinematics, then relativistic dynamics, lastly energy.

Definition of physics:

Definition 1 Physics is the natural quantitative science that deals with the fundamental interactions between matter and energy.

What arguably sets physics apart from the other sciences would be the reductionist nature. At the base of the "pyramid of the natural sciences" lies mathematics, the fundamental subjects. Up a level is Physics, then Chemistry, then Biology. Because it's quantitative, when we study the world around us with measurements and standardizations. This introduces us to *dimensions and unit*:

Claim 1 Nearly all properties in classical physics can be described with one or more of the three dimensions: length, time, and mass.

The notation would be $[x]$ as the dimension of some variable or constant, for example

$$[x] = L. \quad (1.1)$$

The SI unit is the international unit. Remember that the standard for mass in SI is kilograms. Now we introduce the natural unit.

Definition 2 The natural unit is defined by setting the values of the three constants:

$$c = \hbar = G = 1. \quad (1.2)$$

Here they still have dimensions, for example:

$$[c] = \frac{L}{T} = 1 \quad \Rightarrow \quad L = T. \quad (1.3)$$

Dimensional Analysis

A simple pendulum on Earth has length L , mass m . Given

$$[l] = L, [m] = M, \text{ and } [g] = \frac{L}{T^2} \quad (1.4)$$

determine, up to dimensionless factors, the period of the pendulum.

Solution. We have, apparently, the period τ with dimension T . We would assume the

powers with greek letters:

$$[\tau] = T = [l^\alpha m^\beta g^\gamma] = L^\alpha M^\beta \left(\frac{L}{T^2} \right)^\gamma. \quad (1.5)$$

This is solved simply as:

$$\begin{cases} \alpha + \gamma = 0, \\ \beta = 0, \\ -2\gamma = 1 \end{cases} \Rightarrow \begin{cases} \alpha = \frac{1}{2}, \\ \beta = 0, \\ \gamma = -\frac{1}{2} \end{cases}. \quad (1.6)$$

Which gives us

$$\tau \propto l^{\frac{1}{2}} g^{-\frac{1}{2}} = \sqrt{\frac{l}{g}} \quad (1.7)$$

Exercise 1 Look at the dimensions of c , $\hbar$, and G . Create the Planck units- the Planck length, time, and mass, by combining the three constants in a unique way that they create a constant for length, time, and mass.

SECTION 2

Definitions

We start by defining the basics of kinematics- the study of "how things move". The simplest case of motion is none- an object being stationary. Now there will be a question: where is it. Now we define the frame of reference and point particles.

Definition 3 A frame of reference for an observer is a sufficiently precise laboratory of measurement approaches, i.e. clocks, meter sticks. Here, in classical physics, we assume a precise measurement without quantum fluctuation, and it can be arbitrarily precise.

And we define the point particle:

Definition 4 A point particle is an approximation whereby an object is treated as a zero-dimensional, mathematical point. It is valid when the size and shape of the object are irrelevant for the question under study.

In this frame of reference, we may set up Cartesian coordinates for space (x, y, z) to uniquely specify the position of a particle. The Cartesian coordinate is defined by three everywhere mutually perpendicular axes intersecting at the same point, defined as the origin.

Now it's the problem of incorporating time: the sequence of locations a particle passes can be described as time. It cannot, to our best knowledge be reversed, so it's the monotonically increasing parameter denoting the particle's position. We treat it as another label on the event: write (ct, x, y, z) . Thus, to locate an event in space, we have four dimensions.

Remark 1 Notice that the time axis is not t but ct , because the dimension of it should be in length.

Thus we have the more formal definition or general statement:

Claim 2 To specify an event in the universe, we provide four labels:

$$(ct, x, y, z) \quad (2.1)$$

where x, y and z are for space or location, and ct as time. Here c is introduced as a speed constant (not "light speed" in its literal sense), which we will cover in due course.

This encourages us to think in the perspective that the "arena" of the universe is a four-dimensional mathematical space that we will call spacetime. A teaser for the mathematical space: "manifolds". This will be covered in higher division courses.

Now we have the empirical facts about space.

Claim 3 Here are the empirical facts about space:

1. It is in three dimensions, with an extra dimension of time,
2. It is isotropic, which means it looks the same in every direction,
3. It is homogenous, as in it is the same at every point.

In the larger context of general relativity, which includes gravity, there are many space-times. To distinguish ours, the one relevant for special relativity, we will call it Minkowski Spacetime. Thus to summarize, ***The universe is a four dimensional Minkowski spacetime, with points and events labeled (ct, x, y, z) .***

Observers and inertial frames. There is a special class of frames of reference are known as inertial frames. It is the result of Newton's first law, which we will cover later in the course.

Consider a free particle— a particle subject to no influences. (we assume that this is true in the context of classical mechanics). Then, an inertial frame is a frame in which a free particle moves uniformly in a straight line, or not at all. This implies that if one inertial frame exists, by moving the origin or setting another frame of reference that is moving at constant speed relative to our first, or even performing static rotations, these will all define inertial frames equally.

A straight line in this sense can be defined as the shortest distance between two points in space— when we talk about geometric spaces, we are talking about the geometry of Euclid. Another remark is that coordinates are entirely artificial, and the choice of coordinates is arbitrary. Thus the laws of physics *cannot* depend on coordinates. This actually has profound consequences: the laws that we write down are not arbitrary and are subject to the principle.

Take the vector for example. It is defined as a "quantity with a magnitude and a direction", independent of coordinate systems— that is the motivation for vectors, scalars, tensors... and other ideas in physics.

SECTION 3

Coordinate Systems

Distance. To express distances in coordinate systems, we have Pythagorean's Theorem:

Theorem 1 For two points in space, in a Euclidean coordinate system, we have the distance, $\Delta\ell$ is defined as:

$$\Delta\ell^2 = \Delta x^2 + \Delta y^2 + \Delta z^2. \quad (3.1)$$

If the quantity is infinitesimal:

$$d\ell^2 = dx^2 + dy^2 + dz^2. \quad (3.2)$$

Remark 2 Note that using the calculus of variations would reveal the straight line as the shortest distance between two arbitrary points.

Henceforth— and unless otherwise stated— we will look in inertial frames. The first we will introduce, of course, is the Cartesian coordinates, where we choose three mutually perpendicular axes. The other two coordinate systems, spherical / cylindrical coordinates are defined well enough in Calculus, with the exception of using (ct, r, θ, φ) where θ is the azimuthal angle and φ is the polar coordinate, different from the mathematical definition.

Now we introduce the shorthand notation for the axes:

$$(ct, x, y, z) \Rightarrow (x^0, x^1, x^2, x^3). \quad (3.3)$$

Or, when in spacetime coordinates we have:

$$x^\mu, x^\nu, x^\lambda, x^\sigma, \mu, \nu, \lambda, \sigma \in [0, 3], \quad (3.4)$$

or, when we are in space,

$$x^i, x^j, x^m, x^k, i, j, m, k \in [1, 3]. \quad (3.5)$$

There will be an abuse of notation that is quite standard:

$$x^\mu = (x^0, x^1, x^2, x^3). \quad (3.6)$$

Defining time. The natural choice of parameterization is to use coordinate time:

$$t \equiv \frac{x^0}{c}. \quad (3.7)$$

Then, we would have

$$x^0 = x^0(t) = ct \Rightarrow x^\mu = x^\mu(t) = (ct, x^1(t), x^2(t), x^3(t)). \quad (3.8)$$

When we use Cartesian coordinates, or $x^1 = x, x^2 = y, x^3 = z$, we would have the positions defined, when we suppress the time factor for simplicity, would be

$$(x(t), y(t), z(t)). \quad (3.9)$$

Now we move on to the definition of a "frame", or a "slice" of spacetime, as:

Definition 5 A frame is a spatial slice of spacetime where every point on the slice is simultaneous with every other.

We can consider this as a "stack of slices", where the progression sequence is the direction of time. But this is inaccurate: it has the underlying assumption that space and time are independent of observers: what is considered simultaneous in the perspective of one observer is not necessarily simultaneous in another. We will come back to this notion in time dilation.

Displacement. As a first attempt at motion, define the displacement Δx^i as:

$$\Delta x^i \equiv x_f^i - x_i^i. \quad (3.10)$$

Where $x_f^i = x^i(t_f)$ and $x_i^i = x^i(t_i)$, where the duration is $\Delta t = t_f - t_i$. We can immediately see that this is a crude definition of motion because it considers only the endpoint and starting point, providing just the "Line of Sight" distance between two points. So, to further clarify, we will define also its magnitude:

$$|\Delta x^i| \equiv \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}. \quad (3.11)$$

We notice that this is an extremely crude way still, because it does not count for any motion of the particle in between the time of motion. However if we take the limit, this becomes extremely useful. Let's first obtain this from the Euclidean metric.

PROOF We first introduce a parameter, λ , where the line is parameterized by

$$x(\lambda) = a\lambda, \quad y(\lambda) = b\lambda, \quad z(\lambda) = c\lambda. \quad (3.12)$$

With $0 \leq \lambda \leq 1$. Let

$$L = |\Delta x^i|, \quad (3.13)$$

This, when we start with a line assuming $c = 0$, we get $y = \frac{b}{a}x$. When we take the length differential,

$$d\ell = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}, \quad (3.14)$$

the length would be:

$$L = \int d\ell = \int_0^1 \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2} d\lambda = \sqrt{a^2 + b^2 + c^2}. \quad (3.15)$$

For this context is a bit of an overkill, but for spacetime the incorporation of time would lead to more counterintuitive answers. $\square$

SECTION 4

Vectors in Three-Dimensional Euclidean Space

From the definition of displacement we notice that it is correlated with direction- it has both a magnitude, and a direction. Thus we introduce the idea of vectors:

Definition 6 A quantity with only magnitude is a scalar. A quantity with magnitude and direction is a vector. A vector can thus be denoted as a directed line segment, i.e. an arrow. Note that vectors are independent of coordinates.

Now we go on to define the algebra, transformation, and calculus of vectors. We will start with notation. Vectors will be noted as $\vec{A}$ or $\mathbf{A}$, with its magnitude $|\vec{A}|$ or A . For the direction of the vector we will denote as the unit vector $\hat{\mathbf{A}}$.

In Cartesian coordinates this course will prefer $\{\hat{x}, \hat{y}, \hat{z}\}$. But there is a better and more general way to note this:

$$\{\hat{e}_1, \hat{e}_2, \hat{e}_3\}. \quad (4.1)$$

This extends to the claim:

Claim 4 Any arbitrary vector $\hat{\mathbf{A}}$ can be expressed as a linear combination of the basis vectors $\{\hat{e}_i\}$.

Vector addition. Geometrically we define the diagonal of the parallelogram formed by the two vectors to be added. We also define multiplication by a scalar:

Definition 7 Let $\alpha \in \mathbb{R}$ and $\hat{e}_i$ as the basis vectors. Define $\alpha \hat{e}^i$ by:

$$|\alpha \hat{e}_i| = |\alpha|. \quad (4.2)$$

If $\alpha > 0$ the result is parallel to the original vector, if $\alpha < 0$ is antiparallel to the original vector, if $\alpha = 0$ the result is a zero vector.

We also define, formally, vector addition as the diagonal of the parallelogram formed by the two vector components. This is important because we can now express any vector using these three base vectors:

$$\mathbf{A} = A^1 \hat{e}_1 + A^2 \hat{e}_2 + A^3 \hat{e}_3. \quad (4.3)$$

Where the components of $\mathbf{A}$ are $\{A^1, A^2, A^3\}$, and they can be zero or nonzero. Here the components are in superscript because in abstract algebra (and by extension four-dimensional space), these will represent component vectors. The magnitude of $\mathbf{A}$ is given by

$$|\mathbf{A}| \equiv A = \sqrt{(A^1)^2 + (A^2)^2 + (A^3)^2}. \quad (4.4)$$

We can now write, for two arbitrary vectors $\mathbf{A}$ and $\mathbf{B}$, and scalars, α and β , we can write the following:

Definition 8 Linear combinations:

$$\alpha \mathbf{A} + \beta \mathbf{B} = (\alpha A^1 + \beta B^1) \hat{e}_1 + (\alpha A^2 + \beta B^2) \hat{e}_2 + (\alpha A^3 + \beta B^3) \hat{e}_3. \quad (4.5)$$

*Note that we do not need to provide definitions for subtraction because $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$. There is, though, a useful way of drawing the geometric- the final vector minus the initial vector, linking the tips of $\mathbf{A}$ and $\mathbf{B}$.

This sounds like a mouthful and is complicated to write. Usually we write out the sigma in a compact form, with i from 1 through 3, but it is still too difficult. Note, though, that i is a *dummy variable*, meaning that within the summation the variables can be reassigned for other letters. They are essentially placeholders for numbers, so in any sum we can freely change the letter i . But, if the variables are not summed (when they are *free variables*), they are not interchangeable. So, with that in mind, we introduce

The Einstein Summation Convention. We are writing the following:

Claim 5 We write the vector $\mathbf{A}$ as:

$$\mathbf{A} = A^1 \hat{\mathbf{e}}_1 + A^2 \hat{\mathbf{e}}_2 + A^3 \hat{\mathbf{e}}_3 = \sum_{i=1}^3 A^i \hat{\mathbf{e}}_i \equiv A^i \hat{\mathbf{e}}_i. \quad (4.6)$$

Essentially we omit the sigma. It should be one superscript and one subscript, however in Euclidean space ($\mathbb{E}^3$) this does not matter; thus we could write $A_i \hat{\mathbf{e}}_i$. However in the Minkowski space this is important. Now we introduce an example:

Example 1 Write the following using the summation convention:

1. $\mathbf{A} \pm \mathbf{B}$,
2. $\alpha \mathbf{A}$.

Solution. (a) The first one is easily deduced:

$$\alpha \mathbf{A} + \beta \mathbf{B} = (\alpha A^1 + \beta B^1) \hat{\mathbf{e}}_1 + (\alpha A^2 + \beta B^2) \hat{\mathbf{e}}_2 + (\alpha A^3 + \beta B^3) \hat{\mathbf{e}}_3 \quad (4.7)$$

Writing out the summation and dropping the sigma:

$$\mathbf{A} \pm \mathbf{B} = (A^i \pm B^i) \hat{\mathbf{e}}_i. \quad (4.8)$$

(b) the second one we already did:

$$\alpha \mathbf{A} = \alpha A^i \hat{\mathbf{e}}_i. \quad (4.9)$$

Now an exercise question:

Exercise 2 Write A and $\hat{\mathbf{A}}$ using the summation convention.

Solution. We first write out the intermediate equation:

$$A = \sqrt{(A^1)^2 + (A^2)^2 + (A^3)^2}, \quad (4.10)$$

and it would change into (for clarity we don't put extra superscripts):

$$A = \sqrt{A^i A^i}. \quad (4.11)$$

And the unit vector $\hat{\mathbf{A}}$ would be easy:

$$\hat{\mathbf{A}} = \frac{A^i \hat{\mathbf{e}}_i}{\sqrt{A^i A^i}}. \quad (4.12)$$

Scalar products / Dot products. We define an operation (the "product") that multiplies vectors and gives scalars. This is noticed as $\mathbf{A} \cdot \mathbf{B}$. It is defined as:

Definition 9 The dot product is defined as:

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta, \quad (4.13)$$

where θ is the smallest angle between A and B . Note that it is symmetric and commutative, where $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$.

This introduces us to the Kronecker Delta:

$$\hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_j = \delta_{ij}, \quad (4.14)$$

where the delta is a compact form of the identity matrix.

$$\delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases} \quad (4.15)$$

This means we have an orthonormal basis. Now we prove that $\mathbf{A} \cdot \mathbf{B} = A^i B^i$.

PROOF We use the Kronecker Delta:

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= (A^i \hat{\mathbf{e}}_i) \cdot (B^j \hat{\mathbf{e}}_j) = A^i B^j \delta_{ij} \\ &= \sum_{i=1}^3 \sum_{j=1}^3 A^i B^j \delta_{ij} \\ &= A^i B^i. \end{aligned} \quad (4.16)$$

□

Now we go for an exercise:

Exercise 3

Show that:

1. $\mathbf{A} \cdot \mathbf{A} = A^i A^i = A^2$,
2. $\hat{\mathbf{e}}_i \cdot \mathbf{A} = A^i$.

Solution. (a) For the first equation, we simply plug in $\mathbf{A}$ for the general definition above and have:

$$\mathbf{A} \cdot \mathbf{A} = A^i A^i. \quad (4.17)$$

And, because we know, from Equation 4.11,

$$A = \sqrt{A^i A^i} \Rightarrow A^2 = A^i A^i, \quad (4.18)$$

we can prove the original equation.

(b) We expand the equation:

$$\hat{\mathbf{e}}_i \cdot \mathbf{A} = \hat{\mathbf{e}}_i \cdot (A^j \hat{\mathbf{e}}_j) = A^j \delta_{ij} = A^i. \quad (4.19)$$

Another example here: we know that the Euclidean line element

$$d\ell^2 = dx^2 + dy^2 + dz^2. \quad (4.20)$$

Example 2

Show that:

$$d\ell^2 = \delta_{ij} dx^i dx^j. \quad (4.21)$$

Solution. We have:

$$\begin{aligned}
 d\ell^2 &= dx^2 + dy^2 + dz^2 \\
 &= (dx^1)^2 + (dx^2)^2 + (dx^3)^2 \\
 &= \sum_{i=1}^3 (dx^i)^2 = \sum_{i=1}^3 dx^i dx^i \\
 &= \sum_{i=1}^3 \sum_{j=1}^3 \delta_{ij} dx^i dx^j.
 \end{aligned} \tag{4.22}$$

Drop the summation and we get the result. This is why δ_{ij} is called the Euclidean metric.

An extension of this example is this: Suppose

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu, \tag{4.23}$$

solving for g gives you the function for gravity, which is the definition for "geometry of space". Also when we define

$$ds^2 \equiv \left(1 - \frac{2M}{r}\right) c^2 dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 \sin^2 \theta d\Omega^2, \tag{4.24}$$

we can determine the event horizon of a black hole because when $r = 2M$, the second differential gets to infinity.

We now define the **Cross product**.

Definition 10 Multiply two vectors and get a vector through $\mathbf{A} \times \mathbf{B}$. We have:

$$\mathbf{A} \times \mathbf{B} = AB \sin \theta \hat{\mathbf{n}}, \tag{4.25}$$

where the direction of $\hat{\mathbf{n}}$ is defined by the right hand rule. Note that this is anti-symmetric product (thus evidently not associative), because by the right hand rule,

$$\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}. \tag{4.26}$$

Now we need to write down the cross product of the basis vectors. We have:

$$\hat{\mathbf{e}}_i \times \hat{\mathbf{e}}_j = \epsilon_{ijk} \hat{\mathbf{e}}_k. \tag{4.27}$$

The ϵ is defined as the Levi-Civita symbol, where

$$\epsilon_{ijk} = \begin{cases} 1, & \text{if } \{i, j, k\} \text{ is an even permutation of } \{1, 2, 3\}, \\ -1, & \text{if } \{i, j, k\} \text{ is an odd permutation of } \{1, 2, 3\}, \\ 0, & \text{otherwise.} \end{cases} \tag{4.28}$$

Note that k is a free variable. The exercise now is to expand the vector multiplication of the base vectors.

Exercise 4 The first one is

$$\hat{\mathbf{e}}_1 \times \hat{\mathbf{e}}_2 = \epsilon_{121} \hat{\mathbf{e}}_1 + \epsilon_{122} \hat{\mathbf{e}}_2 + \epsilon_{123} \hat{\mathbf{e}}_3 = \epsilon_{123} \hat{\mathbf{e}}_3 = \hat{\mathbf{e}}_3, \tag{4.29}$$

The second one is:

$$\hat{\mathbf{e}}_2 \times \hat{\mathbf{e}}_3 = \epsilon_{231}\hat{\mathbf{e}}_1 + \epsilon_{232}\hat{\mathbf{e}}_2 + \epsilon_{233}\hat{\mathbf{e}}_3 = \epsilon_{231}\hat{\mathbf{e}}_1 = \hat{\mathbf{e}}_1, \quad (4.30)$$

And the last one is:

$$\hat{\mathbf{e}}_1 \times \hat{\mathbf{e}}_3 = \epsilon_{132}\hat{\mathbf{e}}_2 = -\hat{\mathbf{e}}_2. \quad (4.31)$$

Now consider two arbitrary vectors $\mathbf{A}$ and $\mathbf{B}$. We have:

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= (A^i \hat{\mathbf{e}}_i) \times (B^j \hat{\mathbf{e}}_j) \\ &= A^i B^j (\hat{\mathbf{e}}_i \times \hat{\mathbf{e}}_j) = A^i B^j (\epsilon_{ijk} \hat{\mathbf{e}}_k) \\ &= \epsilon_{ijk} A^i B^j \hat{\mathbf{e}}_k \\ &= \epsilon_{123} A^1 B^2 \hat{\mathbf{e}}_3 + \epsilon_{132} A^1 B^3 \hat{\mathbf{e}}_2 + \epsilon_{213} A^2 B^1 \hat{\mathbf{e}}_3 \\ &\quad + \epsilon_{231} A^2 B^3 \hat{\mathbf{e}}_1 + \epsilon_{312} A^3 B^1 \hat{\mathbf{e}}_2 + \epsilon_{321} A^3 B^2 \hat{\mathbf{e}}_1 \\ &= (A^2 B^3 - A^3 B^2) \hat{\mathbf{e}}_1 + (A^3 B^1 - A^1 B^3) \hat{\mathbf{e}}_2 + (A^1 B^2 - A^2 B^1) \hat{\mathbf{e}}_3. \end{aligned} \quad (4.32)$$

This is cleaner than the traditional determinant format because it is reproducible in higher dimensions. Note though, that only the Levi-Civita symbol can be defined, and in four dimensions or more, we cannot define the cross product comfortably because there is one more dimension of ambiguity. Now we have an example.

Example 3 | Show that the identity holds:

$$\epsilon_{ijk} \epsilon_{kmn} = \delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}. \quad (4.33)$$

| We know that for this the single free variable is k .

Also: Prove the identity

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}. \quad (4.34)$$

Change of basis. Our basis vectors are coordinate basis vectors, derived from the coordinates we choose. Hence, if we change the coordinate, it implies a change of basis. The components would change after changes of basis, however, the dimensions of the vector itself never changes. We now discuss the change of coordinates through (a) a static rotation, and changing Cartesian to polar coordinates, both in 2D.

We start with rotation. Suppose we have the basic 2D Cartesian coordinates $\mathcal{S}$. Now we rotate it by an angle θ and get $\bar{\mathcal{S}}$. What we need is to get how are the coordinates $(\bar{x}, \bar{y})$ related to (x, y) ? First, from the figure, we have:

$$\begin{aligned} \bar{x} &= x \cos \theta + y \sin \theta, \\ \bar{y} &= -x \sin \theta + y \cos \theta. \end{aligned} \quad (4.35)$$

Sidenote: this raises an immediate question: since distance is preserved, what are the coordinate transformations that preserve the Euclidean line element— that is,

$$d\bar{l}^2 = dl^2? \quad (4.36)$$

Clearly, any $(\bar{x}, \bar{y}) = (x + a, y + b)$ where a and b are constants preserves dl^2 . This

restricts us to linear transformations:

$$\begin{aligned}\bar{x} &= Ax + By, \\ \bar{y} &= Cx + Dy.\end{aligned}\tag{4.37}$$

To prove the uniformity of $d\bar{l}^2$:

$$\begin{aligned}d\bar{l}^2 &= d\bar{x}^2 + d\bar{y}^2 \\ &= (A^2 + C^2)dx^2 + (B^2 + D^2)dy^2 + 2(AB + CD)dxdy \\ &= dx^2 + dy^2.\end{aligned}\tag{4.38}$$

This gives us:

$$\begin{aligned}A^2 + C^2 &= 1, \\ B^2 + D^2 &= 1, \\ AB + CD &= 0.\end{aligned}\tag{4.39}$$

Solve the equation and we can get the matrix in the equation below (4.40):

$$\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix} = x \cdot \begin{pmatrix} \cos \theta \\ -\sin \theta \end{pmatrix} + y \cdot \begin{pmatrix} \sin \theta \\ \cos \theta \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \cdot \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.\tag{4.40}$$

The basis vectors will change as any vector in the coordinate. Thus we can write a shorthand of:

$$\hat{\mathbf{e}}_{\bar{i}} = \hat{\mathbf{e}}_j R_{j\bar{i}}.\tag{4.41}$$

For any vector $\mathbf{A}$, we must have $\bar{\mathbf{A}} = \mathbf{A}$. Hence, we would need:

$$\bar{\mathbf{A}} = \mathbf{A}^{\bar{i}} \hat{\mathbf{e}}_{\bar{i}} = \mathbf{A}^{\bar{i}} (\hat{\mathbf{e}}_j R_{j\bar{i}}) \hat{\mathbf{e}}_j \equiv \mathbf{A}^j \hat{\mathbf{e}}_j = \mathbf{A}.\tag{4.42}$$

This implies that:

$$\mathbf{A}^{\bar{i}} R_{j\bar{i}} = \mathbf{A}^j.\tag{4.43}$$

If we would like to get $\mathbf{A}^{\bar{i}}$ from $\mathbf{A}^j$ we would like an inverse that has the property

$$R_{\bar{k}j} R_{j\bar{i}} = \delta_{\bar{k}\bar{i}}.\tag{4.44}$$

This would imply that

$$\mathbf{A}^{\bar{i}} R_{\bar{k}j} R_{j\bar{i}} = \mathbf{A}^{\bar{i}} \delta_{\bar{k}\bar{i}} = \mathbf{A}^{\bar{k}} = R_{\bar{k}j} \mathbf{A}^j.\tag{4.45}$$

This introduces the rotation matrix, $\mathcal{R}$, where

$$\begin{aligned}(\mathcal{R})_{kj} &= R_{\bar{k}j} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \\ (\mathcal{R}^{-1})_{kj} &= R_{j\bar{k}} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.\end{aligned}\tag{4.46}$$

Note that $\mathcal{R}^{-1} = \mathcal{R}^T$ and thus $\mathcal{R}^{-1}\mathcal{R} = \mathcal{R}^T\mathcal{R} = \mathbb{I}$, and that $\det(\mathcal{R}) = 1$. The set of 2×2 matrices for which $\mathcal{R}^{-1} = \mathcal{R}^T$ and $\det(\mathcal{R}) = 1$ is known as the Special Orthogonal Group, $SO(2)$. An element of the set is called a rotation matrix because its operation on vectors is to rotate them through a fixed angle θ . Now an example:

Exercise 5

Consider dx^i as the components of a vector dx . Show, using rotational matrices $(\mathcal{R}_{\bar{j}\bar{i}})$, that $d\bar{\ell}^2 = d\ell^2$.

Solution. We expand $d\bar{\ell}^2$:

$$\begin{aligned}
 d\bar{\ell}^2 &= \delta_{\bar{i}\bar{j}} dx^{\bar{i}} dx^{\bar{j}} \\
 &= \delta_{\bar{i}\bar{j}} (\mathcal{R}_{\bar{i}m} dx^m) (\mathcal{R}_{\bar{j}n} dx^n) \\
 &= \delta_{\bar{i}\bar{j}} \mathcal{R}_{\bar{i}m} \mathcal{R}_{\bar{j}n} dx^m dx^n \\
 &= \mathcal{R}_{\bar{i}m} \delta_{\bar{i}\bar{j}} \mathcal{R}_{\bar{j}n} dx^m dx^n \\
 &= \mathcal{R}_{\bar{i}m} \mathcal{R}_{\bar{i}n} dx^m dx^n \\
 &= (\mathcal{R}^T)_{m\bar{i}} (\mathcal{R})_{\bar{i}n} dx^m dx^n \\
 &= \delta_{mn} dx^m dx^n = d\ell^2.
 \end{aligned} \tag{4.47}$$

Polar coordinates. We express polar coordinates with (r, θ) . The definitions are well-known, but note that there is a *coordinate singularity* at the origin: when $r = 0$, θ can be any value. This, when we change back to Cartesian coordinates, will disappear, and we know that this is not a physical singularity. This will be studied more in general relativity (4.24).

The polar coordinates are effectively rotating θ throughout $\theta \in [0, 2\pi]$. This results in the definitions of the basis vectors to be related to the value of θ :

$$\begin{aligned}
 \hat{e}_r(\theta) &= \hat{e}_x \cos \theta + \hat{e}_y \sin \theta, \\
 \hat{e}_\theta(\theta) &= -\hat{e}_x \sin \theta + \hat{e}_y \cos \theta.
 \end{aligned} \tag{4.48}$$

While calculating the derivatives- we need to keep in mind when we take derivatives.

SECTION 5

Vector Calculus

We define a time dependent vector, $\mathbf{A} = \mathbf{A}(t)$. Now we derive its derivative with respect to time. We claim that:

$$\frac{d\mathbf{A}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{A}(t + \Delta t) - \mathbf{A}(t)}{\Delta t}. \tag{5.1}$$

When we talk about "calculus with vectors", the important distinction is the calculus of time dependent vs position dependent ones ($\mathbf{A}(t)$ vs $\mathbf{A}(x, y, z)$). We will shift to position dependent ones later on. We go back to the proof- geometrically the vectors would be simple.

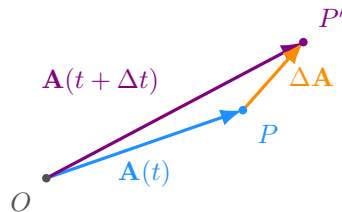


Figure 1. Geometric interpretation of vector derivatives.

We can picture the change as two parts: the parallel part that tracks the change in the magnitude of $\mathbf{A}$, and the perpendicular part that tracks the change in direction of the unit vector $\hat{\mathbf{A}}$. Thus, we would have different cases:

$$\left(\frac{d\mathbf{A}}{dt}\right)_{\perp} = 0 \quad \Rightarrow \quad \frac{d\mathbf{A}}{dt} = \left(\frac{d\mathbf{A}}{dt}\right)_{\parallel}. \quad (5.2)$$

If we have $\mathbf{A}(t) = A(t)\hat{\mathbf{A}}$, then:

$$\frac{d\mathbf{A}}{dt} = \frac{dA}{dt}\hat{\mathbf{A}} + A\frac{d\hat{\mathbf{A}}}{dt} = \left(\frac{d\mathbf{A}}{dt}\right)_{\parallel}. \quad (5.3)$$

Another condition is when the parallel component is zero. We have:

$$\frac{d\mathbf{A}}{dt} = \left(\frac{d\mathbf{A}}{dt}\right)_{\perp}. \quad (5.4)$$

From the Leibniz Rule, we have:

$$\frac{d}{dt}(\vec{\mathbf{A}} \cdot \hat{\mathbf{A}}) = \frac{d}{dt}(\mathbf{A}) = 0. \quad (5.5)$$

Thus,

$$\frac{d\mathbf{A}}{dt} = \frac{dA}{dt}\hat{\mathbf{A}} + A\frac{d\hat{\mathbf{A}}}{dt} = \left(\frac{d\mathbf{A}}{dt}\right)_{\parallel} + \left(\frac{d\mathbf{A}}{dt}\right)_{\perp}. \quad (5.6)$$

This would lead to two components of $d\mathbf{A}/dt$, and the result would be that differentiating a vector is differentiating each of its components. That is,

Definition 11 The derivative of a vector in $\mathbb{E}^3$, with the limit definition, is:

$$\frac{d\mathbf{A}}{dt} = \frac{dA^x}{dt}\hat{\mathbf{e}}_x + \frac{dA^y}{dt}\hat{\mathbf{e}}_y + \frac{dA^z}{dt}\hat{\mathbf{e}}_z. \quad (5.7)$$

Integration is equally straightforward:

$$\int \mathbf{A}(t)dt = \hat{\mathbf{e}}_x \int A^x(t)dt + \hat{\mathbf{e}}_y \int A^y(t)dt + \hat{\mathbf{e}}_z \int A^z(t)dt. \quad (5.8)$$

But what if we are calculating in polar basis? We have:

$$\mathbf{A}(t) = A^r\hat{\mathbf{e}}_r + A^\theta\hat{\mathbf{e}}_\theta. \quad (5.9)$$

Recall that the basis vectors are also rotating.

$$\begin{aligned} \frac{d\hat{\mathbf{e}}_r}{dt} &= \dot{\theta}\hat{\mathbf{e}}_\theta, \\ \frac{d\hat{\mathbf{e}}_\theta}{dt} &= -\dot{\theta}\hat{\mathbf{e}}_r. \end{aligned} \quad (5.10)$$

This can be translated into uniform circular motion:

$$\begin{aligned} \mathbf{v} &= v\hat{\mathbf{e}}_\theta, \\ \mathbf{v}' = \mathbf{a} &= v\frac{d\hat{\mathbf{e}}_\theta}{dt} = -v\dot{\theta}\hat{\mathbf{e}}_\theta = -v\omega\hat{\mathbf{e}}_\theta. \end{aligned} \quad (5.11)$$

The velocity in polar coordinates is:

$$\begin{aligned}\mathbf{v}(t) &= \frac{d\mathbf{r}(t)}{dt} = \frac{dr}{dt}\hat{\mathbf{e}}_r + r(t)\frac{d\hat{\mathbf{e}}_r}{dt} \\ &= \dot{r}\hat{\mathbf{e}}_r + r\dot{\theta}\hat{\mathbf{e}}_\theta.\end{aligned}\tag{5.12}$$

Acceleration is thus:

$$\begin{aligned}\mathbf{a}(t) &= \frac{d\mathbf{v}(t)}{dt} = \ddot{r}\hat{\mathbf{e}}_r + \dot{r}\frac{d\hat{\mathbf{e}}_r}{dt} + (r\ddot{\theta} + \dot{r}\dot{\theta})\hat{\mathbf{e}}_\theta + r\dot{\theta}\frac{d\hat{\mathbf{e}}_\theta}{dt} \\ &= \ddot{r}\hat{\mathbf{e}}_r + \dot{r}\dot{\theta}\hat{\mathbf{e}}_\theta + r\ddot{\theta}\hat{\mathbf{e}}_\theta + \dot{r}\dot{\theta}\hat{\mathbf{e}}_\theta - r\dot{\theta}^2\hat{\mathbf{e}}_r \\ &= (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{e}}_r + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\mathbf{e}}_\theta.\end{aligned}\tag{5.13}$$

Note that we are using both Newtonian and Leibniz notation for clarity, stacking hats and dots are not optimal. For integration it is best practice to convert to Cartesian coordinates and integrate.

SECTION 6

Kinematics in Euclidean Space

We now discuss simple kinematics in $\mathbb{E}^3$ in index notation. A particle position is defined as:

Definition 12

A particle's position is given, at coordinate time t , in a particular inertial frame $((ct, x, y, z) = (x^0, x^1, x^2, x^3))$ by

$$(x^1(t), x^2(t), x^3(t)).\tag{6.1}$$

Here, an observer has used their coordinate time as a parameter to track the particle's changing position.

We define the displacement, Δx^i , over the duration Δt by:

Definition 13

The displacement is defined as:

$$\Delta x^i = x_f^i - x_0^i,\tag{6.2}$$

Where

$$x_f^i = x^i(t_f), \quad x_0^i = x^i(t_0).\tag{6.3}$$

This will involve the definition of the position "vector" (which is not a mathematical vector because it changes if the coordinate system changes e.g. displacement of the origin):

$$\mathbf{x} = x^i\hat{\mathbf{e}}_i.\tag{6.4}$$

The Cartesian vector is just substituting x, y, z for i . Now we define the displacement vector for Cartesian coordinates:

$$\Delta \mathbf{x} = \mathbf{x}_f - \mathbf{x}_0 = \Delta x\hat{\mathbf{e}}_x + \Delta y\hat{\mathbf{e}}_y + \Delta z\hat{\mathbf{e}}_z = \Delta x^i\hat{\mathbf{e}}_i.\tag{6.5}$$

The physical path of curve in space the particle moves along is called a particle's trajectory. This is NOT x vs t . Over a finite duration Δt_1 , the displacement vector in general does not accurately describe the motion of the particle. We can do better by using dis-

placements over shorter time intervals. In fact, if we take the limit of $\Delta t \rightarrow 0 \equiv dt$, then $\Delta \mathbf{x}$ approaches the accurate path the particle took. We have:

$$\mathbf{x}(t + dt) = \mathbf{x}(t) + d\mathbf{x}(t). \quad (6.6)$$

Moreover, $|d\mathbf{x}| = d\ell$ the length of the particle's trajectory. We expect $d\mathbf{x} \propto dt$ because we expect the particle's trajectory in the interval to approach a straight line. We need a vector here:

$$d\mathbf{x} = \mathbf{v}dt. \quad (6.7)$$

Thus we introduce the rigor of calculus to write:

Claim 6 The velocity of a particle is defined as:

$$|\mathbf{v}| = \lim_{\Delta t \rightarrow 0} \frac{\Delta \ell}{\Delta t} = \frac{d\ell}{dt}, \quad (6.8)$$

And:

$$\mathbf{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{x}}{\Delta t} = \frac{d\mathbf{x}}{dt} = \dot{\mathbf{x}}. \quad (6.9)$$

Here, $\mathbf{v}$ is called the velocity vector of the particle. Its direction is the instantaneous direction of the particle's motion; it is always tangent to the particle's trajectory. The magnitude of $\mathbf{v}$ is called the speed of the particle along its motion, where speed is our quantitative measure of the quickness of motion.

Thus we have the equation for the conversion from velocity to position. Suppose $\mathbf{v}(t) = \mathbf{v}_0$ as a constant vector:

$$\int_0^t \frac{d\mathbf{x}}{dt'} dt' = \int_0^t \mathbf{v}_0 dt' \Rightarrow \mathbf{x}(t) = \mathbf{x}_0 + \mathbf{v}_0 t. \quad (6.10)$$

The same logic is applied to acceleration:

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \dot{\mathbf{v}}(t) = \ddot{\mathbf{x}}(t). \quad (6.11)$$

Now we do an example.

Exercise 6

A particle's trajectory is a circle of radius R in the xy plane, centered at the origin. It revolves around the circle at a uniform rate, that is, the angular speed $\omega = \dot{\theta}$ is constant, where θ is the angle the particle's position vector makes with the positive x -axis. Find $\mathbf{x}(t)$, $\mathbf{v}(t)$, $\mathbf{a}(t)$.

Solution.

The central problem of classical mechanics is: given an acceleration vector $\mathbf{a}(t)$, we need to find its position. The formal solution is also simple: Given $\mathbf{a}(t)$, $\mathbf{v}_0$, and $\mathbf{x}_0$:

$$\begin{aligned} \mathbf{a} = \frac{d\mathbf{v}}{dt} &\Rightarrow \int_0^t \mathbf{a}(t') dt' = \mathbf{v}(t) - \mathbf{v}_0 \Rightarrow \mathbf{v}(t) = \mathbf{v}_0 + \int_0^t \mathbf{a}(t') dt', \\ \mathbf{v} = \frac{d\mathbf{x}}{dt} &\Rightarrow \int_0^t \mathbf{v}(t') dt' = \mathbf{x}(t) - \mathbf{x}_0 \Rightarrow \mathbf{x}(t) = \mathbf{x}_0 + \mathbf{v}_0 t + \int_0^t \int_0^{t'} \mathbf{a}(t'') dt'' dt'. \end{aligned} \quad (6.12)$$

We now do an example.

Exercise 7

A particle's acceleration is given by the equation $\mathbf{a}(t) = a_0 e^{-\alpha t} \hat{\mathbf{e}}_z$, where a_0 and α are positive constants. Find $\mathbf{v}(t)$ and $\mathbf{x}(t)$ assuming $\mathbf{v}_0 = 0$ and $\mathbf{x}_0 = h \hat{\mathbf{e}}_z$ ($h > 0$).

Solution. the motion is one-dimensional. Since $a^x = a^y = 0$ and $v_0^x = v_0^y = 0$, we have: $v^x(t) = v^y(t) = 0$ and hence (since $x_0 = y_0 = 0$), $x(t) = y(t) = 0$. Thus we need only $v^z(t)$ and $z(t)$.

$$v^z(t) = v_0^z + \int_0^t a^z(t') dt' = \int_0^t a_0 e^{-\alpha t'} dt' = -\frac{a_0}{\alpha} [e^{-\alpha t} - 1]. \quad (6.13)$$

Or,

$$v^z(t) = \frac{a_0}{\alpha} (1 - e^{-\alpha t}). \quad (6.14)$$

For velocity, we have:

$$z(t) = z_0 + \int_0^t v^z(t') dt' = h + \frac{a_0}{\alpha} \int_0^t (1 - e^{-\alpha t'}) dt' = h + \frac{a_0}{\alpha} \left[t + \frac{1}{\alpha} (e^{-\alpha t} - 1) \right]. \quad (6.15)$$

Or,

$$z(t) = h + \frac{a_0}{\alpha} t - \frac{a_0}{\alpha^2} (1 - e^{-\alpha t}). \quad (6.16)$$

Here we introduce constant acceleration motion. We now demonstrate it how to eliminate t to obtain an expression involving $\mathbf{v}$, $\mathbf{x}$, and $\mathbf{a}$ only.

Definition 14

The time independent expressions for $\mathbf{x}$, $\mathbf{v}$, and $\mathbf{a}$

We have the equations:

$$\begin{aligned} \mathbf{x}(t) &= \mathbf{x}_0 + \mathbf{v}_0 t + \frac{1}{2} \mathbf{a}_0 t^2, \\ \mathbf{v}(t) &= \mathbf{v}_0 + \mathbf{a}_0 t, \\ \mathbf{a}(t) &= \mathbf{a}_0. \end{aligned} \quad (6.17)$$

Solution. We have:

$$\mathbf{v} - \mathbf{v}_0 = \mathbf{a}_0 t \quad \Rightarrow \quad (\mathbf{v} - \mathbf{v}_0) \cdot (\mathbf{v} - \mathbf{v}_0) = a_0^2 t^2. \quad (6.18)$$

Also,

$$\mathbf{x} - \mathbf{x}_0 - \mathbf{v}_0 t = \frac{1}{2} \mathbf{a}_0 t^2 \quad \Rightarrow \quad 2\mathbf{a}_0 \cdot \Delta \mathbf{x} - 2\mathbf{v}_0 \cdot \mathbf{a}_0 t = a_0^2 t^2 \quad (6.19)$$

Because we have

$$v^2 = v_0^2 + 2\mathbf{a}_0 \cdot \Delta \mathbf{x}, \quad (6.20)$$

the two equations here will become:

$$(\mathbf{v} - \mathbf{v}_0) \cdot (\mathbf{v} - \mathbf{v}_0) = 2\mathbf{a}_0 \cdot \Delta \mathbf{x} - 2\mathbf{v}_0 \cdot (\mathbf{v} - \mathbf{v}_0), \quad (6.21)$$

$$v^2 + v_0^2 - 2\mathbf{v} \cdot \mathbf{v}_0 = 2\mathbf{a}_0 \cdot \Delta \mathbf{x} - 2\mathbf{v} \cdot \mathbf{v}_0 + 2v_0^2. \quad (6.22)$$

Hence,

$$v^2 = v_0^2 + 2\mathbf{a}_0 \cdot \Delta \mathbf{x}. \quad (6.23)$$

Free fall and projectile motion

Claim 7 All objects, regardless of shape, size, composition, etc., have the same free-fall motion: constant acceleration downward. Here, free fall motion is the movement resulting from the sole influence of Earth's gravity. We denote this acceleration as $g = 9.8\text{m/s}^2$. Looking ahead we can call g the gravitational field, denoted by $\mathbf{g}$ where $\mathbf{g} = -g\hat{\mathbf{e}}_z$, so that $\mathbf{a} = \mathbf{g}$.

SECTION 7

Relativity

Galilean Relativity. We have now understood how to communicate and translate between different frames of reference. The perspective so far is to take a single observer, when they choose a different coordinate. There is an alternative but equivalent way: consider two observers, and the coordinate transformations allowed between S and $\bar{S}$ would be:

$$x^{\bar{i}} = R^{\bar{i}}_j x^j. \quad (7.1)$$

Special Relativity. We begin with a claim:

Claim 8 The principle of relativity: the laws of physics are the same in all inertial frames of reference.

The principle of covariance. The laws of physics must be expressed in terms of quantities that are independent of coordinates, that is, can be unambiguously coordinated or transferred between different frames.

Lorentz Transformations. Two inertial frames, S and $\bar{S}$, with coordinates x^i and $x^{\bar{i}}$. Assert that:

$$ds^2 \equiv -c^2 dt^2 + dx^2 = d\bar{s}^2. \quad (7.2)$$

Now we have the lorentz boosts in the $x^0 - x^1$ plane:

$$\begin{aligned} x^{\bar{0}} &= \gamma (x^0 - \beta x^1) & \Rightarrow & \quad \bar{t} = \gamma \left(t - \frac{V}{c^2} x \right), \\ x^{\bar{1}} &= \gamma (x^1 - \beta x^0) & \Rightarrow & \quad \bar{x} = \gamma (x - Vt), \end{aligned} \quad (7.3)$$

where $\gamma \equiv (1 - \beta^2)^{-1/2}$, and $\beta \equiv v/c$.

Write these as:

$$x^{\bar{\mu}} = \Lambda^{\bar{\mu}}_{\nu} x^{\nu}, \quad (7.4)$$

where $\Lambda^{\bar{\mu}}_{\nu}$ are the elements of the matrix Λ :

$$\Lambda^{\bar{\mu}}_{\nu} = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (7.5)$$

Now we do an example on the relativity of simultaneity.

Exercise 8

Consider our setup. Consider two events, A and B, that are simultaneous in S . That is, $\Delta t_{AB} \equiv t_B - t_A = 0$. They occur at different position $\Delta x_{AB} \neq 0$. What is $\Delta \bar{t}_{AB}$?

Solution. The Lorentz boost solves it straightforwardly:

$$\Delta \bar{t}_{AB} = \gamma \left(\Delta t_{AB} - \frac{V}{c^2} \Delta x_{AB} \right) = -\gamma \frac{V}{c^2} \Delta x_{AB} \neq 0. \quad (7.6)$$

SECTION 8

Relativistic Dynamics

Definition of the general dot product:

$$\mathbf{A} \cdot \mathbf{B} = \frac{1}{4} [(A + B)^2 - (A - B)^2]. \quad (8.1)$$

The dual vector space V^* is the set of all linear maps or functions on the vector... (finish here later)

Newtonian laws 1 and 2 hold in Minkowski space. Here, the first law or the law of inertia holds for four-vectors:

$$\frac{du^\mu}{d\tau} = 0. \quad (8.2)$$

The second law is also true: the change in motion is proportional to the motive force impressed and directed along the line in which the force is impressed. But, the third law does NOT hold because the definition of causality is limited to the speed of light / causality.